

LORAIN COUNTY REGIONAL, OH, USA

WMO: 725214

Lat: 41.346N

Lon: 82.179W

Elev: 794

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	7.7	-6.1	4.1	3.3	-0.6	5.5	9.2	28.3	33.3	26.4	30.0	8.4	250	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	89.8	73.4	87.5	72.3	84.4	70.8	76.2	85.4	74.6	83.1	73.1	81.4	10.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	126.7	81.6	72.1	122.2	79.9	70.4	115.1	78.2	40.1	85.1	38.5	83.2	37.2	81.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.7	19.2	DB	-4.3	93.3	6.2	2.5	-8.8	95.1	-12.4	96.5	-15.9	97.9	-20.4	99.7	(4)
			WB	-4.9	78.8	6.1	2.0	-9.3	80.2	-12.9	81.4	-16.3	82.5	-20.7	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	26.9	29.1	37.6	49.8	61.0	69.6	72.8	70.9	65.1	53.9	42.9	32.6
	DBStd	18.56	11.73	11.22	10.89	9.58	8.63	6.87	5.47	5.14	6.93	8.67	9.44	10.15
	HDD50	2709	719	591	411	120	10	0	0	0	1	56	255	546
	HDD65	5851	1182	1006	850	466	186	35	4	10	82	359	665	1006
	CDD50	3118	2	5	27	113	349	587	708	648	454	178	41	6
	CDD65	782	0	0	1	9	60	172	247	192	86	15	0	0
	CDH74	6593	0	0	8	97	580	1419	2173	1481	714	118	3	0
	CDH80	1977	0	0	1	14	161	464	727	397	195	18	0	0
Wind	WSAvg	8.9	11.3	10.7	10.4	10.4	8.7	7.8	6.6	5.9	6.4	8.4	10.0	10.7
Precipitation	PrecAvg	37.1	2.3	2.0	2.7	3.5	3.9	3.8	3.9	3.4	3.4	2.4	3.1	2.7
	PrecMax	48.3	6.1	4.8	5.9	9.0	10.4	7.9	8.5	8.0	9.2	5.7	7.4	8.0
	PrecMin	19.3	0.4	0.2	0.6	0.6	0.6	0.9	1.1	0.6	0.6	0.5	0.7	1.1
	PrecStd	5.6	1.3	1.1	1.1	1.5	2.0	1.5	1.7	1.8	1.9	1.1	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.7	65.8	72.8	82.1	88.4	91.3	93.1	90.4	90.3	82.2	71.8	62.6
		MCWB	55.1	56.5	59.8	64.4	70.2	72.5	75.9	74.8	72.7	68.4	58.8	55.8
	2%	DB	54.5	56.7	65.8	76.7	84.3	88.7	90.0	87.4	85.4	77.3	65.4	56.6
		MCWB	50.5	50.5	55.8	60.7	67.8	72.2	74.3	73.4	71.0	64.7	57.5	51.5
	5%	DB	48.1	50.5	61.0	72.0	81.1	85.5	87.2	84.2	81.4	72.6	61.7	51.7
		MCWB	44.2	45.4	53.4	59.2	66.7	70.6	72.9	72.1	68.9	62.1	53.9	47.3
	10%	DB	43.1	44.6	55.1	66.3	76.4	81.9	84.0	81.6	78.1	68.3	57.3	46.4
		MCWB	38.9	39.6	48.5	55.8	64.3	69.2	71.5	70.7	67.1	59.9	50.9	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	57.2	61.8	67.1	73.2	76.3	79.1	78.0	75.3	70.6	63.0	57.0
		MCDB	59.8	64.3	71.1	77.3	84.5	86.6	90.2	87.0	85.5	80.8	68.1	61.3
	2%	WB	50.9	51.5	57.8	63.5	70.4	74.2	76.9	75.5	73.1	66.8	58.7	52.5
		MCDB	54.1	55.4	64.3	73.6	80.9	84.4	86.9	83.2	81.9	74.2	64.2	56.1
	5%	WB	44.8	45.4	53.9	60.3	68.4	72.5	74.8	74.0	70.9	63.9	55.3	47.3
		MCDB	48.0	49.8	60.5	69.8	77.8	82.4	83.2	81.5	78.6	70.4	60.5	50.4
	10%	WB	39.1	39.5	49.1	57.1	65.9	70.9	73.2	72.4	68.8	60.9	51.7	43.3
		MCDB	42.8	44.1	54.9	65.4	74.4	79.7	81.1	79.3	75.8	67.1	56.9	46.7
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.4	15.6	18.9	19.5	18.3	18.6	18.6	19.8	17.7	14.9	11.9
		MCDBR	18.4	21.8	24.1	26.3	24.5	22.5	21.2	21.4	23.9	24.0	21.0	17.2
		MCWBR	16.0	17.7	16.5	15.7	12.4	10.3	9.5	10.8	12.2	13.7	15.0	14.6
	5% WB	MCDBR	17.8	20.7	22.3	23.7	21.3	19.3	18.1	17.8	19.7	20.6	18.6	16.9
		MCWBR	16.1	17.8	16.0	16.0	12.2	10.2	9.3	10.0	11.0	12.8	14.7	15.7
Clear-Sky Solar Irradiance	taub	0.305	0.334	0.356	0.393	0.428	0.445	0.443	0.429	0.383	0.365	0.328	0.311	
	taud	2.406	2.289	2.312	2.276	2.215	2.213	2.241	2.266	2.371	2.399	2.466	2.451	
	Ebn at Noon	269	275	282	279	270	265	264	264	270	261	258	256	
	Edh at Noon	25	33	36	41	44	45	43	41	34	30	24	23	
All-Sky Solar Radiation	RadAvg	444	709	1061	1440	1714	1898	1907	1689	1354	878	556	370	
	RadStd	43	95	99	118	144	144	134	114	72	121	72	84	39

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	-1.69	N/A	N/A	N/A	N/A
	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	N/A

Nomenclature: See separate page